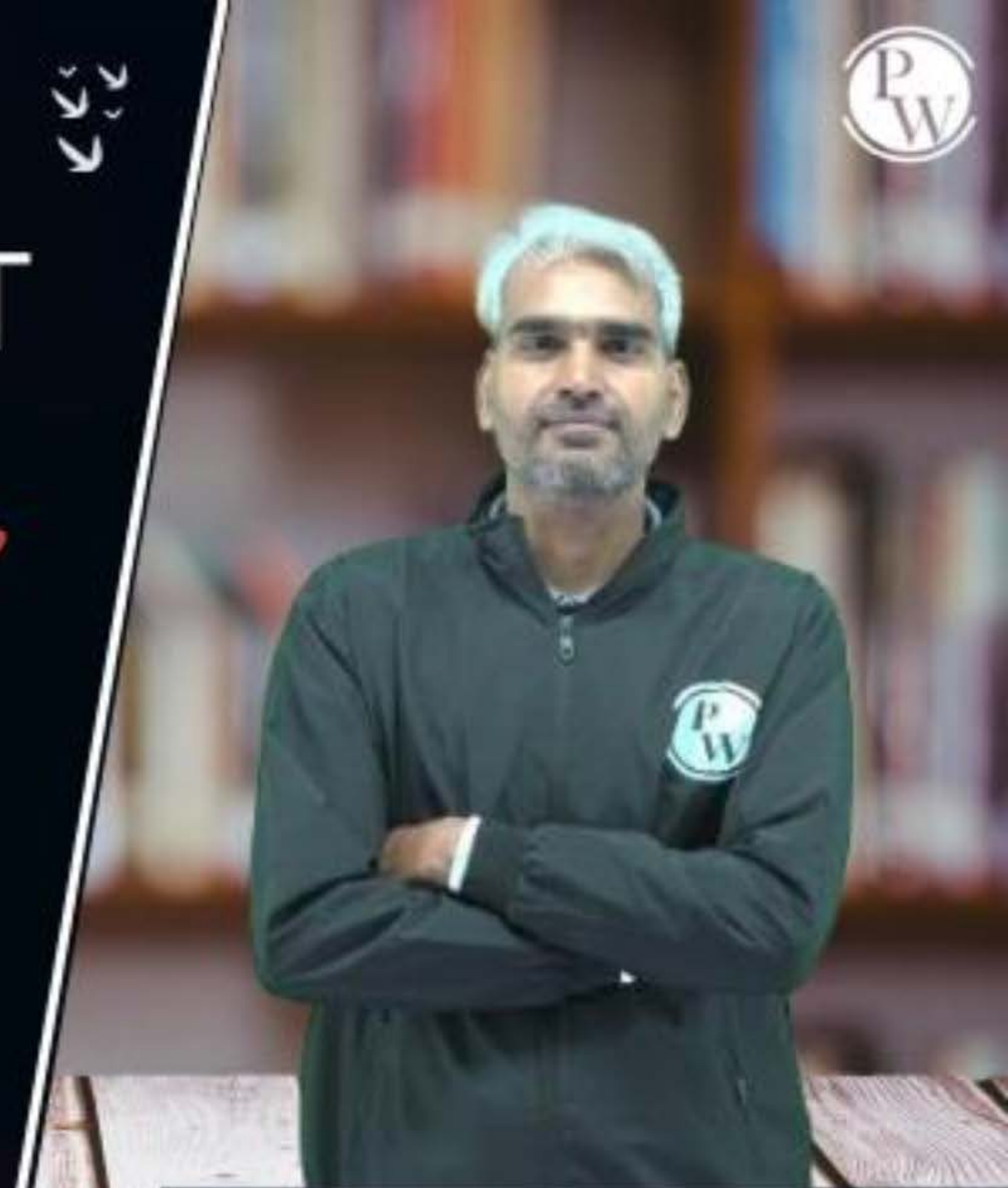
Computer Science & IT

Database Management System

Query Languages

Lecture No. 08



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Recap of Previous Lecture

Topic SQL clauses

Topic Introduction to nested query

Slide

Topics to be Covered

Topic IN, ANY, ALL and EXISTS operators

Topic Execution w.r.t. correlated subquery

Slide

Topic : Nested queries

Sub-query Concept :-

* Retrieve Sids of all students from Student table who scored maximum marks.

it will select the maximum marks from Complete Student table

```
Select Sid
From Student
Where (Marks = (Select Max(Marks) From Student))
```

Outer query (or) Main query

Inner query (or) Nested query (or) Sub-query

Nested Query (Sub-query)

Independent Nested query (Independent Sub-query)	Corelated Nested query (Corelated sub-query)
• If inner query can be executed independently, then it is called independent nested query	* When execution of inner query requires the value of an attribute from the relation specified in outer query, then it is called Corelated Nested query

Nested Query (Sub-query)

Independent Nested query (Independent Sub-query)

eg: Select S.Sid
From Student S
Where (S.Marks = (Select Max(Marks)
From Student))

it can be executed independently

∴ Independent Nested query

Correlated Nested query (Correlated Sub-query)

eg: Select R.A
From R
Where ..... (Select *
From S
Where (S.B = R.C))

'R' is the relation w.r.t outer query

Rel^n w.r.t inner query

Operator

it can not be executed independently because we need the value of an attribute from relation used in outer query

∴ it is called Correlated query

+ "Order of Execution w.r.t. independent Nested query".

1. Inner query will be executed first and it will produce its output.

and then 2. Outer query will execute it will use the o/p produced by inner query. for its execution

Topic : Independent nested query

```
Select Sid
From Student
Where (Marks = (Select Max(Marks)
                From Student))
```

finally those Sids will be produces as o/p by main query $\Rightarrow$ o/p =

Sid
S3
S4

inner query will execute first, and it will produce o/p = $\boxed{60}$

when outer query executes w.r.t. o/p produced by inner query, then tuples w.r.t. Sid = S_1 and Sid = S_2 are selected

Topic : Correlated nested query

Select *

From R

Where operator (Select "List of attributes"

From S

Where (S.A = R.B))

outer query

inner query

operator → This operators will return True or False

→ If it returns True, then Where Condⁿ is True.

→ If it returns false, then Where Condⁿ is False.

R.B → attribut from outer relⁿ

∴ Correlated query.

(Order of Execution)

Topic : Correlated nested query

V.V.V. Important

⑥ ← Select *

① — From R

⑤ ← Where operator (Select * — ④

From S — ②

Where (S.A = R.B)) — ③

R

-	B	- -	-
☆			
'	'	'	'

1st tuple

2nd tuple

S

A	-	- .
#		

1st tuple

2nd tuple

Last tuple

These three steps will be executed for all tuples of relation 'S' (inner relation) w.r.t. each tuple of relation R (outer relation)

w.r.t. each tuple of outer relation inner query will produce a set of tuples. It may be Empty or May not be Empty.

Topic : Operators

* Best used operators with independent query are IN, ANY or ALL.
* Best used operator with Correlated query is EXISTS.

IN, ANY, ALL and EXISTS Can be used with any type of Subquery i.e. Independent or Correlated

IN operator :→

IN operator is used to check whether the concerned tuple is present in the set of tuples produce by the inner query or not.

Complement of 'IN' is 'NOT IN'

→ Check whether

Set of tuples Produced by inner query

it is the concerned tuple → X IN $\{2, 4, 5, 7, 8\}$ or not

if $X = 5$, then 'IN' operator will return true {as '5' is Present in the set of tuples}

if $X = 6$, then 'IN' operator will return false {as '6' is not Present in the set of tuples}

→ Check whether

X IN $\{(a,1), (b,1), (b,2), (c,3), (d,3)\}$ or not

if $X = (b, 2)$, then IN returns true

if $X = (c, 1)$, then IN returns false.

Note:- If set of tuples is Empty, i.e. when output Produced by inner query is Empty, then "IN" operator will always return false, and hence 'NOT IN' will always return true.

Consider the following relations.

Supplier (Sid, Sname, Rating)

Parts (Pid, Color)

Catalog (Sid, Pid)

Query:- Retrieve Sids of all the suppliers who have supplied some "red" color Parts

S_1 P_1 ✓

S_1 P_2 ✗

S_2 P_2 ✗

S_3 P_2 ✗

S_3 P_3 ✓

```
Select distinct Sid
From Catalog
Where Pid IN (Select Pid
              From Parts
              Where Color = 'Red')
```

O/P = { P_1, P_3 }

It will produce a set that contains all Pids of Red Color Parts

O/P:

Sid
S_1
S_3

Soln 1:-

```
Select distinct C.Sid
From Catalog C, Parts P
Where (C.Pid = P.Pid
       AND
       P.Color = 'RED')
```

Catalog

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2
S_3	P_2
S_3	P_3

Parts

Pid	Color
P_1	Red
P_2	Green
P_3	Red

ANY operator

operator 'ANY' is used along with other $\underbrace{\text{Comparison operators.}}$

i.e; <, >, <=, >=, =, <>

Not equal

- **Note:** "ANY" will return true if and only if at least one tuple in the set of tuples {i.e; in the o/p of inner query} satisfy the Comparison Condition with tuple under Consideration

→ Check whether

x < ANY $\{2, 4, 5, 7, 8, 9\}$ or not

Set of tuples Produced by inner query

it is the concerned tuple

w.r.t. '5' — × × × ✓

$5 < 7$

Condⁿ true ∴ will not compare with remaining tuples

w.r.t. '15' — × × × × × ×

None of tuples in the set of tuples satisfy the comparison condⁿ!

∴ ANY will return false

if $x = 5$, then "ANY" operator will return true

if $x = 15$ then "ANY" operator will return false

Note:- ① If inner query result is Empty then operator "ANY" will always return false.

② "IN" is equivalent to " = ANY".

"ALL" operator

operator 'ALL' is used along with other $\underbrace{\text{Comparison operators}}_{\text{i.e.; } <,\ >,\ <=,\ >=,\ =,\ <>}$.

Not equal

- $\underline{\underline{\text{Note}}}$: "ALL" will return false if and only if at least one tuple in the set of tuples {i.e; in the o/p of inner query} fails the Comparison Condition with 'tuple' under Consideration

→ Check whether

Set of tuples Produced by inner query

it is the concerned tuple → x $>$ALL $\{2, 4, 5, \underbrace{7, 8, 9}\}$ or not

w.r.t. '5' — ✓ ✓ ✗ will not Compare with remaining elements.

└ As soon as an element is found in the set of tuples that fails the Comparison Condⁿ, operator 'ALL' will return false!

w.r.t. '15' → ✓ ✓ ✓ ✓ ✓ ✓ → None of tuples in the set of tuples fails the Comparison Condⁿ! ∴ "ALL" will return true

if $x = 5$, then "ALL" operator will return false

if $x = 15$ then "ALL" operator will return true

Note :- ① If inner query result is Empty then operator "ALL" will always return "true"

② "NOT IN" is equivalent to "<> ALL".

x NOT IN $\{2, 3, 5, 7\}$	x <> ALL $\{2, 3, 5, 7\}$
$x = 4$, then "NOT IN" return true	$x = 4$, then ALL returns true

EXISTS operator

* If inner query result is not Empty, then "EXISTS" will return True
* If inner query result is Empty, then EXISTS will return False.

i.e, "EXISTS return true if and only if inner query result is not Empty"

Note: Complement of EXISTS is "NOT EXISTS"

What output will be produce by the following query On the relations Catalog Parts

#e.g. SELECT C.sid

FROM Catalog C

WHERE EXISTS (SELECT *

FROM Parts P

WHERE (P.Pid=C.Pid AND P.color='RED'))

Catalog

Sid	Pid
S_1	P_1
S_1	P_2
S_3	P_2
S_3	P_3

Parts

Pid	Color
P_1	Red
P_2	Green
P_3	Red

Select C.Sid

from Catalog C

Where EXISTS (Select * $\xRightarrow{o/p}$ $\boxed{P_1 \quad Red}$

from Parts P

Where (C.Pid = P.Pid AND P.Color = 'Red'))

C

1 $\boxed{S_1 \quad P_1}$

Not Empty

∴ EXISTS return True

↓

∴ Where Cond^n is true

∴ 1st tuple of Catalog is selected

Catalog

	Sid	Pid
1	S_1	P_1
2	S_1	P_2
3	S_3	P_2
4	S_3	P_3

→ S_1 P_1 ✓

Parts

Pid	Color	
P_1	Red	✓
P_2	Green	×
P_3	Red	×

② $\boxed{S_1 \quad P_2}$

```
Select C.Sid
  from Catalog C
    Where EXISTS ( Select * {Empty}
                     from Parts P
                     Where (C.Pid = P.Pid AND P.Color = 'Red'))
```

∴ False

↓

Hence 2^{nd} tuple of Catalog is not selected.

Parts

Pid	Color
P_1	Red
P_2	Green
P_3	Red

Catalog

	Sid	Pid
1	S_1	P_1
2	S_1	P_2
3	S_3	P_2
4	S_3	P_3

$S_1 \; P_1$ ✓

$\boxed{S_1 \quad P_2}$ ✗

③ $\boxed{S_3 \; P_2}$

Select C.Sid
from Catalog C
Where EXISTS (Select * {Empty}
from Parts P
Where (C.Pid = P.Pid AND P.Color = 'Red'))

∴ False

∴ 3^{rd} tuple is not selected.

Parts

Pid	Color	
P_1	Red	✗
P_2	Green	✗
P_3	Red	✗

Catalog

	Sid	Pid		
①	S_1	P_1	$S_1 \; P_1$	✓
②	S_1	P_2	$\boxed{S_1 \; P_2}$	✗
③	S_3	P_2	$\boxed{S_3 \; P_2}$	✗
④	S_3	P_3		

Select C.Sid
from Catalog C
Where EXISTS (Select * From Parts P
Where (C.Pid = P.Pid AND P.Color = 'Red'))

{ (P3, Red) }

Not Empty
∴ True
Hence 4th tuple is selected

④ S3 P3

Parts

Pid	Color
P1	Red
P2	Green
P3	Red

Catalog

	Sid	Pid
①	S1	P1
②	S1	P2
③	S3	P2
④	S3	P3

S1 P1 ✓
$\boxed{S_1\ P_2}$ ✗
$\boxed{S_3\ P_2}$ ✗
S3 P3 ✓

⇒ From the selected tuples result Sids in o/p

⇒ ∴ o/p =

Sid
S1
S3

H.W.

#e.g. SELECT C1.sid

FROM Catalog C1

WHERE NOT EXISTS (SELECT P.Pid

FROM Parts P

WHERE NOT EXISTS (SELECT C2.Sid

FROM Catalog C2

WHERE(C2.Pid=P.Pid AND C2.Sid=C1.Sid)))

Catalog

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2

Parts

Pid	Color
P_1	Red
P_2	green

What output is produced by above SQL query:

A. Sids of suppliers who supplied some parts

B. Sids of suppliers who supplied only proper subset of parts from all parts

C. Sids of suppliers who supplied all parts

D. Sids of suppliers who did not supply any part

2 mins Summary

Topic IN, ANY, ALL and EXISTS operators

Topic Execution w.r.t. correlated subquery

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THANK - YOU

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